

DEPARTMENT OF COMPUTER SCIENCE

And Software Engineering



Subject Title

Computer Networks

Project Title

AirNet Company Network Design & Security

Developed by:

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Project Title:

AirNet Company Network Design & Security

Using VLSM, VLANs, VPN & Security Controls

1.Introduction and Networking Requirements

“Secure Inter-Branch Network Architecture Using VLSM, VLAN Segmentation and VPN Connectivity”

AirNet Company is a rapidly expanding Australian enterprise headquartered in Sydney, with planned branch offices in Melbourne, Brisbane, and Hong Kong. The company requires a **scalable, secure, cost-effective, and centrally managed network infrastructure** to support inter-branch communication, internet access, and future growth.

Under **Set 3**, the user requirements for each location are:

Location	Number of Users
Sydney (Head Office)	5000
Melbourne	3200
Brisbane	2500
Hong Kong	2785

The company owns the private IPv4 block **188.42.0.0/16**, which must be subnet using **Variable Length Subnet Masking (VLSM)** to efficiently allocate addresses while allowing future growth.

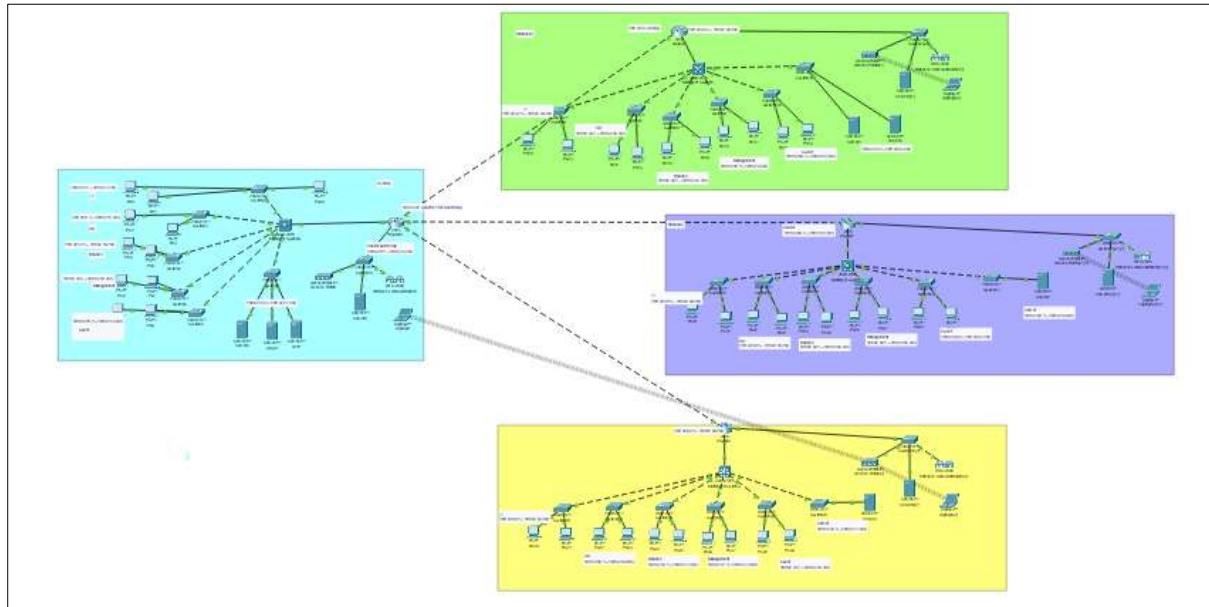
Key Networking Requirements

- Secure inter-branch connectivity between Sydney, Melbourne, Brisbane, and Hong Kong
- Centralised intranet services hosted at Sydney HQ
- Departmental segregation using VLANs
- Dedicated **Server VLAN** on Level 5 of each branch
- Secure wired and wireless access

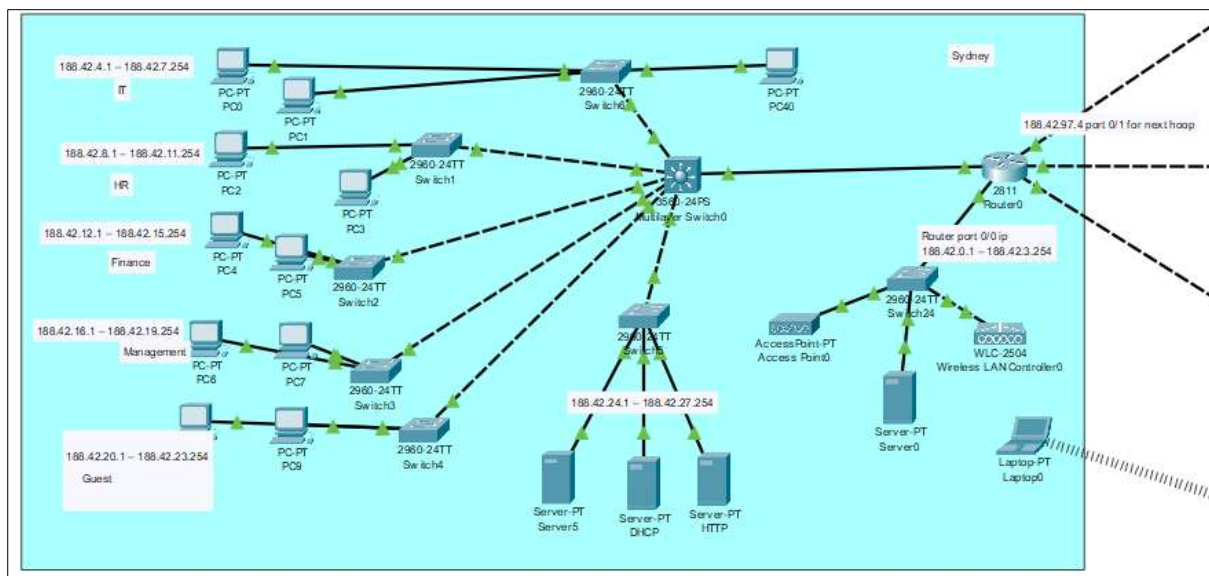
2. Network Topology Design

Chosen Topology: Hierarchical Star Topology

Design:

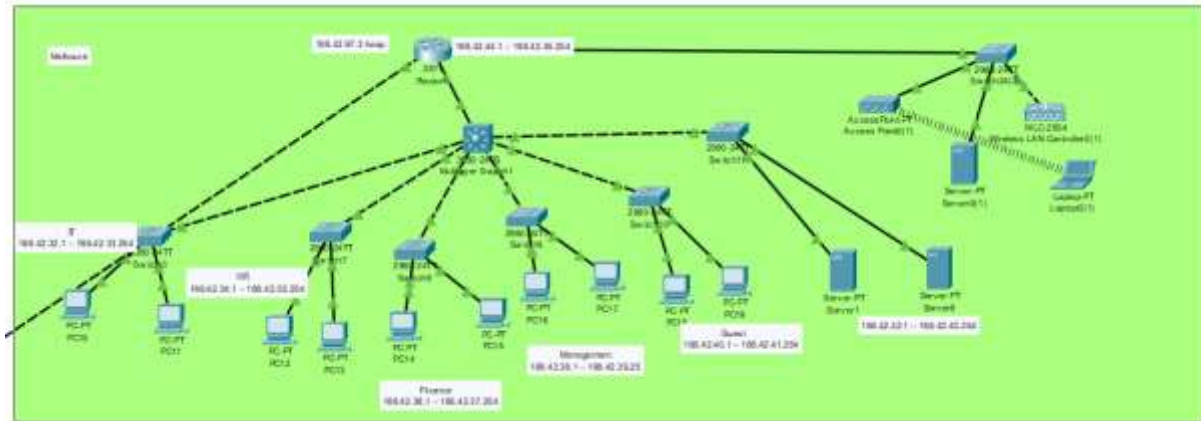


Sydney (Head Quarter)

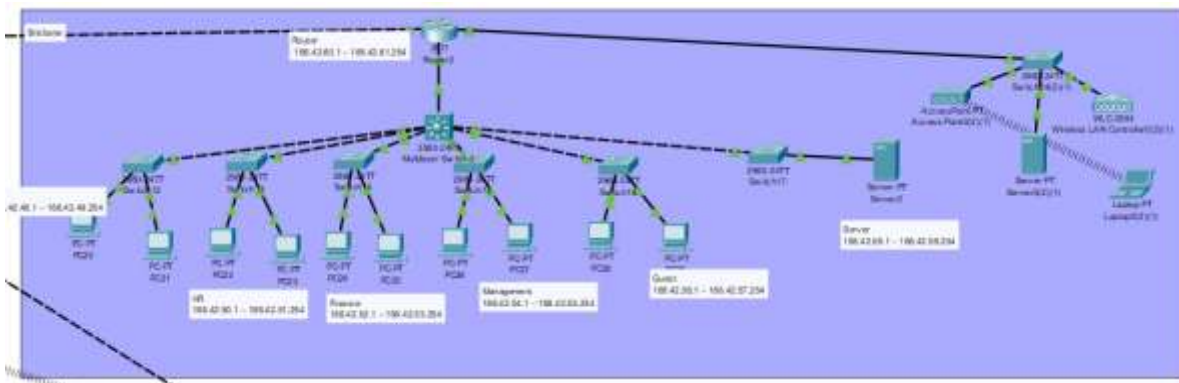


A **hierarchical star topology** was selected as it is widely adopted in enterprise environments and supports scalability, performance, and fault isolation.

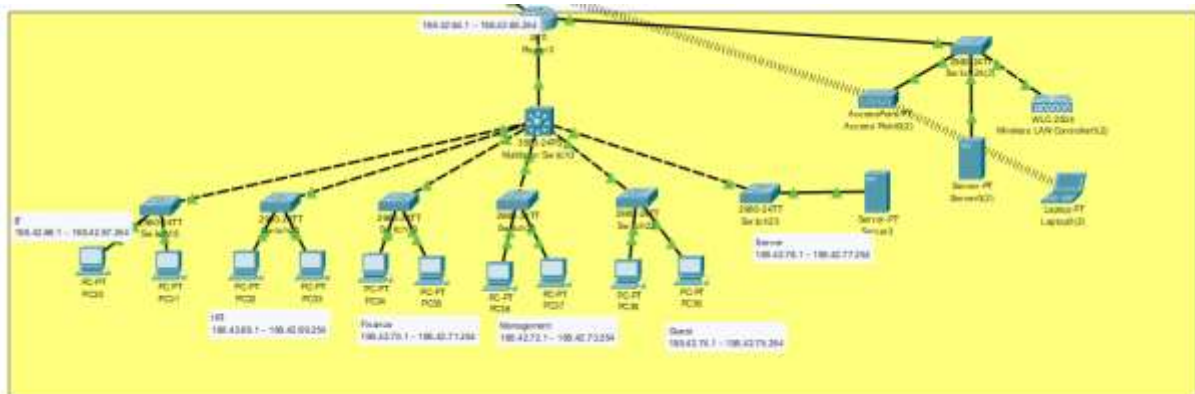
Melbourne



Brisbane



Hong Kong



Topology Structure

- **Core Layer (Sydney HQ)**
 - Enterprise router
 - Next-Generation Firewall
 - VPN termination point
- **Distribution Layer (Each Branch)**

- Layer 3 switches for routing and VLAN segmentation
- **Access Layer**
 - Layer 2 switches on each floor
 - Wireless Access Points

Sydney acts as the **central hub**, while Melbourne, Brisbane, and Hong Kong connect as **spoke sites** through encrypted VPN tunnels. This design ensures centralised management and cost efficiency.

3. Local Area Network (LAN) Design & Packet Tracer Implementation

Each branch building consists of **8 floors**, with **6 servers located in the server room on Level 5**. The LAN design ensures efficient traffic flow and strong security.

LAN Design Features

- Access switches deployed on every floor
- Trunk links connecting access switches to distribution switches
- Inter-VLAN routing via Layer 3 switches
- Dedicated server access switches on Level 5
- Firewall placed between internal LAN and internet

Cisco Packet Tracer was used to:

- Design the network topology
- Configure VLANs and routing
- Assign IP addresses
- Implement VPN tunnels
- Test connectivity and failover

4. IP Addressing, Subnetting and VLSM

The address block **188.42.0.0/16** provides 65,536 usable addresses. Using VLSM, subnets are allocated based on actual host requirements.

Location	Required Hosts	CIDR	Usable Hosts	Reason
Sydney HQ	5000	/19	8190	Supports HQ users and future growth
Melbourne	3200	/20	4094	Adequate for current and future users
Hong Kong	2785	/20	4094	Similar scale to Melbourne
Brisbane	2500	/21	2046*	Efficient allocation using VLAN segmentation

IP Addressing and VLSM Calculations

Base Network: 188.42.0.0/16

1. Base Network Information

- **Network Address:** 188.42.0.0
- **Default Subnet Mask:** 255.255.0.0
- **CIDR Notation:** /16
- **Total IP addresses:**

$$2^{(32-16)} = 2^{16} = 65,536$$

2. VLSM Design Rule

The subnet size is calculated using the formula:

$$H = 2^n - 2 \quad \text{Number of required hosts}$$

Where:

- n = number of host bits
- Subnet mask = $32 - n$

3. Subnet Calculations (Largest to Smallest)

3.1 Sydney HQ (5000 Hosts)

IP Configuration	
☒ DHCP	☐ Static
IPv4 Address	188.42.4.3
Subnet Mask	255.255.252.0
Default Gateway	188.42.4.1
DNS Server	188.42.24.2

IP Configuration	
☒ DHCP	☐ Static
IPv4 Address	188.42.20.3
Subnet Mask	255.255.252.0
Default Gateway	188.42.20.1
DNS Server	188.42.24.2

Calculation

$$2^{13} - 2 = 8190(\text{Sufficient})$$

- **Host bits:** 13
- **Network bits:** $32 - 13 = 19$

Result

- **CIDR:** /19
- **Subnet Mask:** 255.255.224.0
- **Block Size:**

$$256 - 224 = 32$$

Address Range

- **Network Address:** 188.42.0.0
- **First Usable IP:** 188.42.0.1
- **Last Usable IP:** 188.42.31.254

- **Broadcast Address:** 188.42.31.255

3.2 Melbourne (3200 Hosts)

Calculation

$$2^{12} - 2 = 4094(\text{Sufficient})$$

- **Host bits:** 12
- **Network bits:** $32 - 12 = 20$

Result

- **CIDR:** /20
- **Subnet Mask:** 255.255.240.0
- **Block Size:**

$$256 - 240 = 16$$

Address Range

- **Network Address:** 188.42.32.0
- **First Usable IP:** 188.42.32.1
- **Last Usable IP:** 188.42.47.254
- **Broadcast Address:** 188.42.47.255

3.3 Hong Kong (2785 Hosts)

Calculation

$$2^{12} - 2 = 4094(\text{Sufficient})$$

- **Host bits:** 12
- **Network bits:** $32 - 12 = 20$

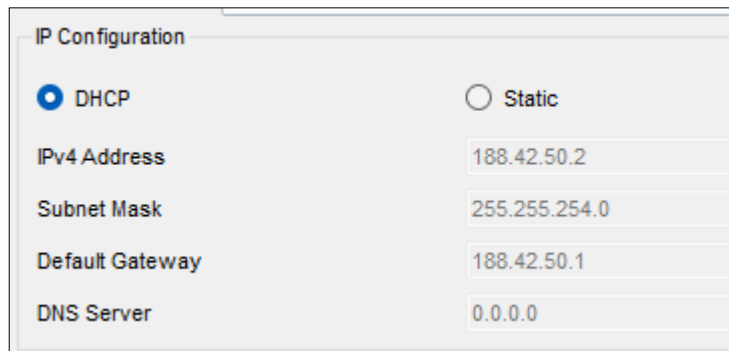
Result

- **CIDR:** /20
- **Subnet Mask:** 255.255.240.0
- **Block Size:** 16

Address Range

- **Network Address:** 188.42.48.0

- **First Usable IP:** 188.42.48.1
- **Last Usable IP:** 188.42.63.254
- **Broadcast Address:** 188.42.63.255



The screenshot shows a network configuration window titled "IP Configuration". It has two tabs: "DHCP" (selected) and "Static". Under the "DHCP" tab, there are five fields with their corresponding values:

Field	Value
IPv4 Address	188.42.50.2
Subnet Mask	255.255.254.0
Default Gateway	188.42.50.1
DNS Server	0.0.0.0

3.4 Brisbane (2500 Hosts)

Calculation

$$2^{12} - 2 = 4094(\text{Sufficient})$$

However, due to VLAN segmentation and cost efficiency, a **/21** subnet is allocated.

- **Host bits:** 11
- **Network bits:** $32 - 11 = 21$

Result

- **CIDR:** /21
- **Subnet Mask:** 255.255.248.0
- **Block Size:**

$$256 - 248 = 8$$

Address Range

- **Network Address:** 188.42.64.0
- **First Usable IP:** 188.42.64.1
- **Last Usable IP:** 188.42.71.254
- **Broadcast Address:** 188.42.71.255

Subnet Allocation Summary

Location	Network Address	CIDR	Subnet Mask	Usable IP Range
Sydney HQ	188.42.0.0	/19	255.255.224.0	188.42.0.1 – 188.42.31.254
Melbourne	188.42.32.0	/20	255.255.240.0	188.42.32.1 – 188.42.47.254
Hong Kong	188.42.48.0	/20	255.255.240.0	188.42.48.1 – 188.42.63.254
Brisbane	188.42.64.0	/21	255.255.248.0	188.42.64.1 – 188.42.71.254

Justification

- /19 provides 8190 hosts for Sydney HQ
- /20 provides 4094 hosts for Melbourne and Hong Kong
- /21 provides 2046 hosts (extended with VLAN segmentation) for Brisbane
- Efficient IP utilisation with room for growth

5. Security Mitigation + Policies + ACS Ethics

5. VLAN Implementation

Departmental VLAN Design

Each branch implements the following VLANs:

VLAN ID	Department
VLAN 10	Management
VLAN 20	Human Resources
VLAN 30	Finance
VLAN 40	IT Support
VLAN 50	Guest Network
VLAN 60	Server VLAN (Level 5)

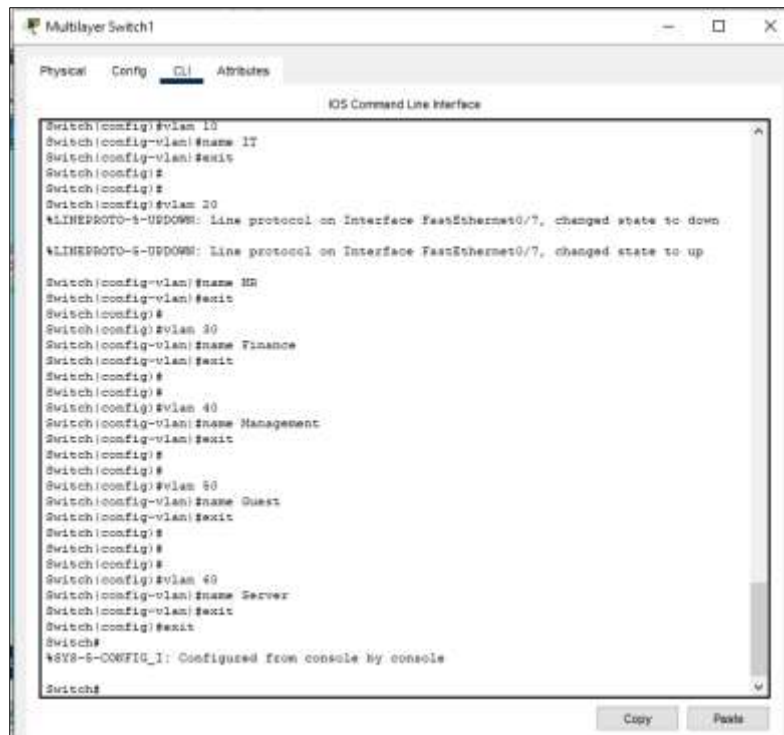
Cisco Packet VLAN implementation:

RIP		VLAN No	VLAN Name
SWITCHING		1	default
VLAN Database		10	IT
INTERFACE		20	HR
FastEthernet0/1		30	Finance
FastEthernet0/2		40	Management
FastEthernet0/3		50	Guest
FastEthernet0/4		60	Server
FastEthernet0/5			

VLAN No	VLAN Name
1	default
10	IT
20	HR
30	Finance
40	Management
50	Guest
60	Server
1002	fddi-default
1003	token-ring-default

Inter-VLAN Routing

- Implemented using **Layer 3 switches**
- **Multi-layer switch:**



Multi-layer switch Vlan Configuration:

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface vlan 10
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
Switch(config-if)#ip address 188.42.32.1 255.255.254.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface vlan 20
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
Switch(config-if)#ip address 188.42.34.1 255.255.254.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#
Switch(config)#interface vlan 30
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up
Switch(config-if)#ip address 188.42.36.1 255.255.254.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#
Switch(config)#interface vlan 40
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan40, changed state to up
```

At the bottom of the window are 'Copy' and 'Paste' buttons.

```

Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#
Switch(config)#interface vlan 40
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan40, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan40, changed state to up

Switch(config-if)#ip address 188.42.38.1 255.255.254.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#interface vlan 50
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan50, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan50, changed state to up

Switch(config-if)#ip address 188.42.40.1 255.255.254.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#
Switch(config)#interface vlan 60
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan60, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan60, changed state to up

Switch(config-if)#ip address 188.42.42.1 255.255.254.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#

```

Copy

Paste

```

Switch(config-if)#ip address 188.42.42.1 255.255.254.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#
Switch(config)#interface vlan 10
Switch(config-if)#ip address 188.42.32.1 255.255.254.0
Switch(config-if)#ip helper-address 188.42.42.2
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#
Switch(config)#interface vlan 20
Switch(config-if)#ip address 188.42.34.1 255.255.254.0
Switch(config-if)#ip helper-address 188.42.42.2
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#
Switch(config)#interface vlan 30
Switch(config-if)#ip address 188.42.36.1 255.255.254.0
Switch(config-if)#ip helper-address 188.42.42.2
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#
Switch(config)#interface vlan 40
Switch(config-if)#ip address 188.42.38.1 255.255.254.0
Switch(config-if)#ip helper-address 188.42.42.2
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#
Switch(config)#interface vlan 50
Switch(config-if)#ip address 188.42.40.1 255.255.254.0
Switch(config-if)#ip helper-address 188.42.42.2

```

Copy

Paste

```

Switch(config)#
Switch(config)#
Switch(config)#interface vlan 50
Switch(config-if)#ip address 188.42.40.1 255.255.254.0
Switch(config-if)#ip helper-address 188.42.42.2
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#

```

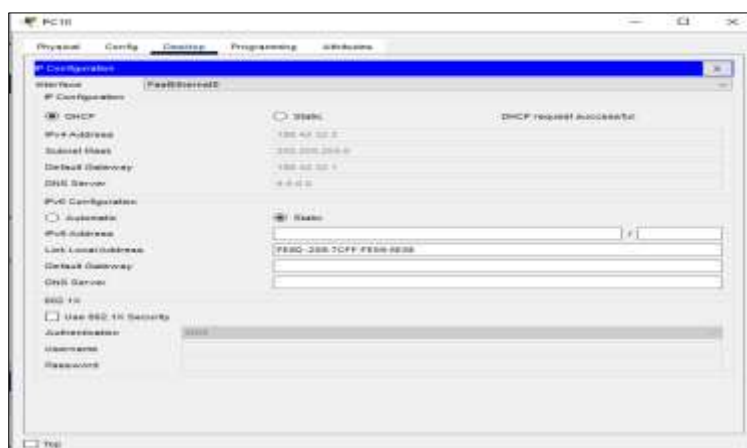
Copy

Paste

Server VLAN

- All servers isolated in VLAN 60
- Only authorised VLANs allowed access
- Enhanced firewall monitoring for server traffic

VLAN	Department	Subnet
VLAN 10	Management	188.42.0.0/24
VLAN 20	HR	188.42.1.0/24
VLAN 30	Finance	188.42.2.0/24
VLAN 40	IT Support	188.42.3.0/24
VLAN 50	Guest	188.42.4.0/24
VLAN 60	Servers	188.42.5.0/24



Vlan 10:

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#interface fa 0/2-3

% Invalid input detected at '^' marker.

Switch(config)#interface range fa 0/2-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#interface fa 0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#exit
Switch(config)#
```

Vlan 20

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#interface range fa 0/2-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#interface fa 0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#exit
Switch(config)#
```

Vlan 30:

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 30
Switch(config-vlan)#name Finance
Switch(config-vlan)#exit
Switch(config)#interface fa 0/2-3

% Invalid input detected at '^' marker.

Switch(config)#interface range fa 0/2-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#interface fa 0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#exit
Switch(config)#
```

Vlan 40:

```

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 40
Switch(config-vlan)#name Management
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#interface range fa 0/2-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#
Switch(config-if-range)#switchport access vlan 40
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#interface fa 0/1
Switch(config-if)#witchport mode trunk
      ^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#exit
Switch(config)#

```

Vlan 50:

```

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 50
Switch(config-vlan)#name Guest
Switch(config-vlan)#exit
Switch(config)#interface range fa0/2-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 50
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#interface fa 0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#exit
Switch(config)#

```

Vlan 60:


```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 60
Switch(config-vlan)#name Server
Switch(config-vlan)#exit
Switch(config)#interface fa 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 60
Switch(config-if)#exit
Switch(config)#
Switch(config)#
Switch(config)#interface fa 0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#exit
Switch(config)#
```

Pinging of Message

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 188.42.32.4

Pinging 188.42.32.4 with 32 bytes of data:

Reply from 188.42.32.4: bytes=32 time=20ms TTL=128
Reply from 188.42.32.4: bytes=32 time<1ms TTL=128
Reply from 188.42.32.4: bytes=32 time<1ms TTL=128
Reply from 188.42.32.4: bytes=32 time<1ms TTL=128

Ping statistics for 188.42.32.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 20ms, Average = 5ms

C:\>|
```

Pinging of message from Melbourne to Brisbane by using IP address

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 188.42.32.16

Pinging 188.42.32.16 with 32 bytes of data:

Reply from 188.42.32.16: bytes=32 time=11ms TTL=128
Reply from 188.42.32.16: bytes=32 time<1ms TTL=128
Reply from 188.42.32.16: bytes=32 time<1ms TTL=128
Reply from 188.42.32.16: bytes=32 time<1ms TTL=128

Ping statistics for 188.42.32.16:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 2ms

C:\>
```

5 (b). Wireless Network (WLAN)

Wireless Deployment

- Wireless Access Points installed on each required floor
- Centralised wireless configuration per site

Wireless Security

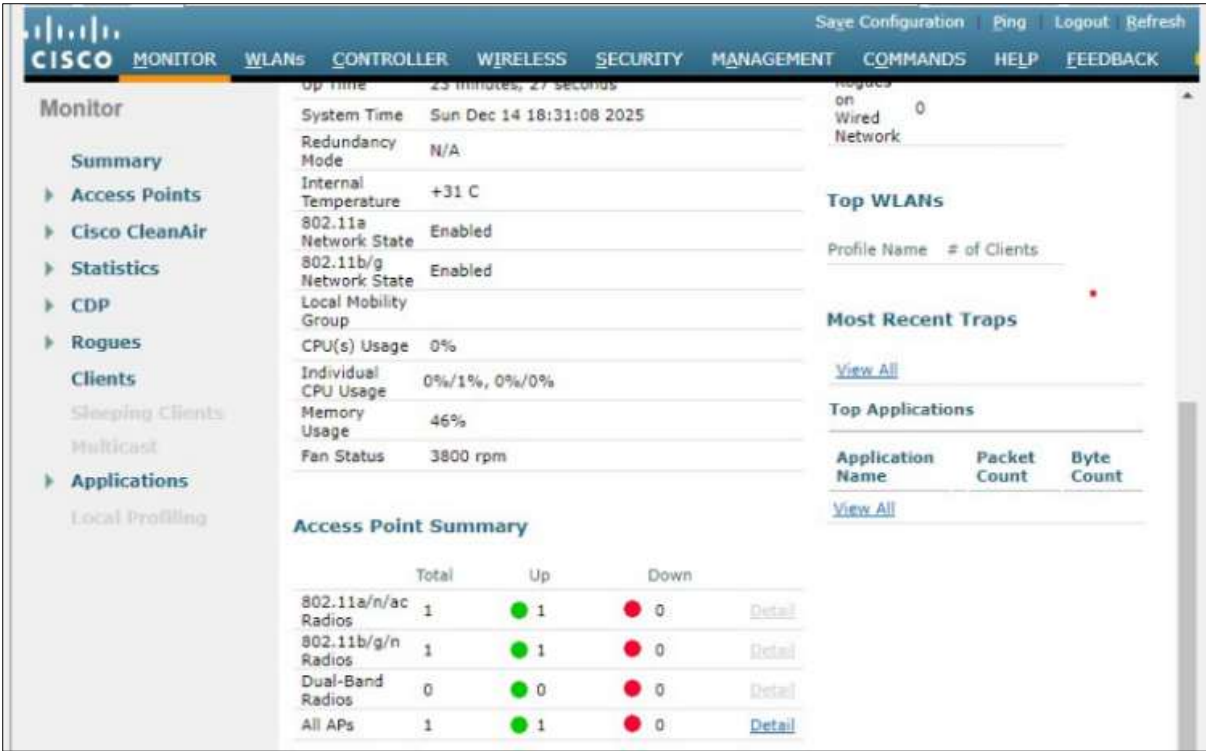
- **Staff Wi-Fi:** WPA3-Enterprise with 802.1X authentication
- **Guest Wi-Fi:** WPA2-PSK with internet-only access

VLAN Mapping

- Staff SSID mapped to internal VLANs
- Guest SSID mapped to VLAN 50

Wireless has been successfully developed and access the point and the link is built successfully.





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6. Hardware and Justification

This section explains the networking hardware used in the AirNet Company network design and justifies each device based on functionality, performance, scalability, security, and cost-effectiveness. All devices are supported in Cisco Packet Tracer and suitable for an enterprise multi-branch environment.

10.1 Enterprise Routers

Role in the Network

- Provide WAN connectivity between HQ and branches
- Perform inter-branch routing
- Act as the internet gateway

Why Cisco ISR Routers

- Support VPN, OSPF
- Packet Tracer supported

Justification

Cisco ISR routers enable secure and cost-effective communication between geographically distributed branches using encrypted VPN connections.

10.2 Layer 3 Switches (Multilayer Switches)

Role in the Network

- Perform inter-VLAN routing
- Act as distribution layer
- Reduce router workload

Why Layer 3 Switches

- High-speed internal routing
- Scalable VLAN design

Justification

Layer 3 switches improve internal network performance by routing traffic locally, reducing latency and supporting a hierarchical network model.

10.3 Layer 2 Access Switches

Role in the Network

- Connect end devices

- Deployed on each floor
- Forward traffic upstream

Why Layer 2 Switches

- Cost-effective
- Support VLANs and trucking

Justification

Layer 2 switches provide efficient and economical access-layer connectivity while maintaining traffic separation through VLANs.

10.5 Wireless Access Points (WAPs)

Role in the Network

- Provide wireless access
- Support staff and guests

Why Wireless Access Points

- Enable mobility
- Support secure wireless standards

Justification

WAPs provide secure and flexible wireless connectivity, with VLAN-based SSIDs ensuring separation between staff and guest access.

10.6 Servers (Level 5 – Server Room)

Role in the Network

- Host file, application, database, and authentication services

Why Dedicated Servers

- Centralized management
- Improved security

Justification

Placing servers in a dedicated Server VLAN ensures controlled access, better performance, and enhanced protection.

10.7 Cabling and Media

Cabling Used

- Cat6 Ethernet

Justification

- Supports high speeds
- Low interference
- Cost-effective and future-ready

7. Conclusion

This project successfully designed and implemented a **secure, scalable, and efficient enterprise network** for AirNet Company using the **188.42.0.0/16** address space. The use of VLSM, VLANs, wireless connections and layered security ensures reliable inter-branch communication, strong protection against cyber threats, and ethical compliance with ACS standards. The design supports future expansion and evolving business needs.

8. Referencing

Cisco Networking Academy (2024)

Australian Computer Society (ACS) – Code of Ethics